

İSMAİL CAN DİKMEN, Ph.D.

Assistant Professor, Department of Electrical and Electronics Engineering
İstinye University, Sarıyer, İstanbul, Türkiye · IEEE Senior Member

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568 citations · h-index 13 · 38 peer-reviewed publications · 3 granted patents · 4 pending patents

Google Scholar, as of July 2026

RESEARCH INTERESTS

- Neuromorphic computing and spiking neural networks
- Intelligent battery management systems
- Hardware–software co-design for edge AI
- Neuromorphic sensing and signal processing

EDUCATION

Ph.D., Electrical and Electronics Engineering — İnönü University, Malatya Thesis: “Design and Application of Smart Battery Management System for Electric Vehicles” Erasmus exchange: Military University of Technology in Warsaw (WAT), Faculty of Cybernetics, Poland.	Sep 2016 – Jun 2022
M.Sc., Electronic Engineering — Aeronautics and Space Technologies Institute, İstanbul Thesis: “Flight Control of a Vertical Take-Off and Landing Aircraft with Four Rotors (Quadrotor)”	Sep 2007 – Aug 2009
B.Sc., Electronic Engineering — Air Force Academy, İstanbul Thesis: “6-Legged Mobile Robot and Applications”	Sep 1997 – Aug 2002

ACADEMIC APPOINTMENTS

İstinye University, İstanbul, Türkiye <i>Assistant Professor, Department of Electrical and Electronics Engineering</i> - Building an application-driven research programme in neuromorphic energy systems and edge AI for safety-critical platforms. - Developing international collaborations and joint research and technology-transfer projects with industry partners.	Jan 2026 – Present
Yıldız Technical University, İstanbul, Türkiye <i>Part-time Lecturer</i> Delivered the Electric Vehicle Technologies course under a Ministry of Industry and Technology / Council of Higher Education programme.	Oct 2024 – Jul 2025
İnönü University, Malatya, Türkiye <i>Head of Department (Electric and Energy), Nov 2021 – Jun 2023 Lecturer, Jun 2020 – Jun 2023</i> - Led departmental strategy, curriculum updates and academic operations; coordinated academic staff and budgets. - Promoted research projects in electric vehicles and energy systems; supported publications and conference participation.	Jun 2020 – Jun 2023
Malatya Turgut Özal University, Malatya, Türkiye <i>Lecturer</i> Contributed to applied R&D activities alongside undergraduate teaching in Electric and Energy programmes.	Jun 2019 – Jun 2020

INDUSTRY AND LEADERSHIP EXPERIENCE

TEMSA Skoda Sabancı Transportation Vehicles Inc., R&D Center, Adana, Türkiye <i>Mobility Technologies Lead (R&D)</i> - Built end-to-end modelling and simulation workflows for product families, reducing test cost, speeding up tender responses and optimising component selection. - Developed BMS algorithms and software optimised for neuromorphic embedded hardware, and took them through ISO 26262 functional safety and regulatory compliance. - Designed and delivered TEMSA’s ZeroTech Academy across Türkiye, Germany and France; evaluation of 315 employees showed a significant increase in electric vehicle knowledge scores from 59.5 to 70.2/100 ($p = 0.019$), with results published in 2025.	Jun 2023 – Feb 2025
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SusTech Research, Development and Consultancy Ltd., Türkiye

Jan 2018 – Jun 2019

Co-Founder

- Developed intelligent BMS solutions for electric vehicles across hardware, embedded software and algorithms.
- Executed the TÜBİTAK-supported E-CAMELEON project (Grant No. 2170454).

MOTAŞ Inc. (Malatya Municipality Public Transportation Operator), Malatya, Türkiye

Dec 2015 – Jan 2017

Lead Engineer / Workshop Engineer

- Managed vehicle procurement, maintenance and repair; prepared technical specifications and production inspections for the trolleybus programme.
- Supervised up to 56 staff and led a team of 8 technicians; established operational standards.

Turkish Air Force

Sep 2002 – Mar 2014

Pilot (Jets and Helicopters), Search and Rescue

- Accumulated 1,029 flight hours across jet and helicopter platforms; executed search-and-rescue missions and operational flights.
- NATO ISAF, Afghanistan (Oct 2011 – May 2012): A-5/7 Plans, Exercise and Evaluation Officer at Kabul International Airport.

RESEARCH PROJECTS**Completed Projects**

- *Comparison, Analysis and Efficiency Optimisation of Virtual and Real Parameters of Electric Vehicle and Welding Technologies through Simulation*. İnönü University Scientific Research Projects (BAP), Project No. FDP-2021-2678. **Researcher**, 2021–2022.
- *Design and Application of a Smart Battery Management System for Electric Vehicles*. İnönü University BAP, Project No. FDK-2021-2645 (doctoral project supporting the Ph.D. thesis). **Researcher**, 2021–2022.
- *Analysis, Modelling and Optimisation for Developing Performance Criteria of the Mechanical, Battery and Electronic Systems of Electric Vehicles*. İnönü University BAP, Project No. FBG-2021-2283. **Researcher**, 2021–2022.
- *Virtual Simulation Laboratory for Electric Vehicle and Welding Technologies*. Firat Development Agency, Project No. TRB1/20/TSG/0035. **Deputy Project Coordinator**, 2020–2021.
- *Orthopaedic Smart Insole*. İnönü University Technopark. **Researcher**, 2019–2021.
- *Modelling and Development of Next-Generation Battery and Power Management Systems for Electric Vehicles, and Analysis of Battery Recycling Processes*. İnönü University BAP, Project No. FOA-2018-1358. **Researcher**, 2018–2020.
- *E-CAMELEON: Adaptive Battery Management System for Electric Vehicles*. TÜBİTAK 1512 (BiGG), Grant No. 2170454, via SusTech R&D. **Researcher**, 2018–2019.

Submitted Proposals — Principal Investigator*TÜBİTAK: The Scientific and Technological Research Council of Türkiye*

- TÜBİTAK 1001, Scientific and Technological Research Projects (2026-1), Project No. 958754. “*Neuromorphic Battery Management for Energy-Critical Systems*” Submitted Mar 2026; under scientific evaluation.
- TÜBİTAK 3501, Career Development Program (CAREER), Project No. 988453. “*Real-Time Radar Micro-Doppler Classification at the Edge with Spiking Neural Networks and Measured Energy-Latency Benchmarking Against Neuromorphic Silicon*” Submitted Jun 2026; under evaluation.

GRANTED PATENTS*All filings with the Turkish Patent and Trademark Office (TÜRKPATENT).*

- [P1] “*Neuromorphic Battery Management System (nBMS)*”, Reg. No. 2025/018585, filed 28 Nov 2025. Inventor: İsmail Can Dikmen. · **Granted**
- [P2] “*Adaptive, Modular, and Intelligent Battery Management System That Can Automatically Determine Battery Chemistry*”, Reg. No. 2021/018973, filed 2 Dec 2021. Assignee: İnönü University. · **Granted**
- [P3] “*An Electronic Method of Determining Battery Chemistry for Lithium-Based Batteries*”, Reg. No. 2021/005464, filed 26 Mar 2021. Assignee: İnönü University. · **Granted**

SELECTED PUBLICATIONS*[†] denotes corresponding author. Journal metrics from JCR 2025 (released June 2026).*

- [1] Dikmen, İ. C.[†] “Low-Energy Battery State Estimation for EVs Using Hybrid SNN-LSTM for Heterogeneous System-on-Chip Architectures”, *IEEE Transactions on Transportation Electrification*, 2026. Early Access, doi:[10.1109/TTE.2026.3707653](https://doi.org/10.1109/TTE.2026.3707653) · SCI-E · Q1 · IF 8.5
- [2] Dikmen, İ. C.[†], Karadağ, T. “Electrical Method for Battery Chemical Composition Determination”, *IEEE Access*, vol. 10, pp. 6496–6504, 2022. doi:[10.1109/ACCESS.2022.3143040](https://doi.org/10.1109/ACCESS.2022.3143040) · SCI-E · Q2 · IF 4.2
- [3] Dikmen, İ. C.[†] “A Physics-Guided Framework for Available Traction Power and In-Motion Charging”, *IEEE Access*, vol. 14, pp. 83959–83970, 2026. doi:[10.1109/ACCESS.2026.3699203](https://doi.org/10.1109/ACCESS.2026.3699203) · SCI-E · Q2 · IF 4.2

- [4] Dikmen, İ. C.[†] “Continuous Monitoring and Modeling of High-Frequency EM Pollution Using Cutting-Edge Regression Techniques”, *IEEE Access*, vol. 14, pp. 46627–46637, 2025. doi:[10.1109/ACCESS.2025.3548841](https://doi.org/10.1109/ACCESS.2025.3548841) · SCI-E · Q2 · IF 4.2
- [5] Dikmen, İ. C.[†], Arısoy, A., Temeltaş, H. “Attitude Control of a Quadrotor”, *2009 4th International Conference on Recent Advances in Space Technologies (RAST)*, pp. 722–727, IEEE, 2009. doi:[10.1109/RAST.2009.5158286](https://doi.org/10.1109/RAST.2009.5158286)
- [6] Dikmen, İ. C.[†] “A Review on Fundamentals, Applications, Challenges and Current Status of Spiking Automotive Electronics”, *IJASTECH*, vol. 10, no. 2, pp. 281–305, 2026. doi:[10.30939/ijastech.1845650](https://doi.org/10.30939/ijastech.1845650) · Scopus · Q3
- [7] Dikmen, İ. C.[†], Yıldırım, N., Karadağ, T. “Machine Learning Approaches for Enhancing the SoH Estimation of LTO Batteries”, *IJASTECH*, vol. 9, no. 1, pp. 48–59, 2025. doi:[10.30939/ijastech.1522403](https://doi.org/10.30939/ijastech.1522403) · Scopus · Q3
- [8] Işık, M., Tiwari, K., Eryılmaz, M. B., Dikmen, İ. C. “Accelerating Sensor Fusion in Neuromorphic Computing: A Case Study on Loihi-2”, *2024 IEEE High Performance Extreme Computing Conference (HPEC)*, pp. 1–7, 2024. doi:[10.1109/HPEC62836.2024.10938501](https://doi.org/10.1109/HPEC62836.2024.10938501)
- [9] Işık, M., Naoukin, J., Dikmen, İ. C. “NeuroVM: Dynamic Neuromorphic Hardware Virtualization”, *2024 IEEE 15th International Green and Sustainable Computing Conference (IGSC)*, pp. 74–79, 2024. doi:[10.1109/IGSC64514.2024.00023](https://doi.org/10.1109/IGSC64514.2024.00023)
- [10] Işık, M., Vishwamith, H., Sur, Y., Inadagbo, K., Dikmen, İ. C. “NEUROSEC: FPGA-Based Neuromorphic Audio Security”, *Applied Reconfigurable Computing (ARC 2024)*, Aveiro, LNCS 14553, pp. 134–147, Springer, 2024. doi:[10.1007/978-3-031-55673-9_10](https://doi.org/10.1007/978-3-031-55673-9_10)
- [11] Yunusoğlu, A., Le, D., Işık, M., Tiwari, K., Dikmen, İ. C., Karadağ, T. “Battery State of Health Estimation Using LLM Framework”, *2025 26th Int. Symposium on Quality Electronic Design (ISQED)*, pp. 1–8, IEEE, 2025. doi:[10.1109/ISQED65160.2025.11014446](https://doi.org/10.1109/ISQED65160.2025.11014446)
- [12] Coşkun, T., Vishwamith, H., Işık, M., Dikmen, İ. C. “Hardware-Accelerated Transformer Framework for Real-Time Battery SoH Estimation”, *2025 IEEE High Performance Extreme Computing Conference (HPEC)*, 2025. doi:[10.1109/HPEC67600.2025.11196577](https://doi.org/10.1109/HPEC67600.2025.11196577)
- [13] Coşkun, T., Vishwamith, H., Işık, M., Naoukin, J., Dikmen, İ. C. “Exploring Neuromorphic Computing with Loihi-2 for High-Performance CFD Simulations”, *2025 IEEE High Performance Extreme Computing Conference (HPEC)*, 2025. doi:[10.1109/HPEC67600.2025.11196267](https://doi.org/10.1109/HPEC67600.2025.11196267)
- [14] Yunusoğlu, A., Le, D., Işık, M., Dikmen, İ. C., Karadağ, T. “Neuromorphic Circuits with Spiking Astrocytes for Increased Energy Efficiency, Fault Tolerance, and Memory Capacitance”, *2025 IEEE 4th International Conference on Computing and Machine Intelligence (ICMI)*, pp. 7–13, 2025. doi:[10.1109/ICMI65310.2025.11141332](https://doi.org/10.1109/ICMI65310.2025.11141332)
- [15] Le, D., Arig, B., Işık, M., Dikmen, İ. C., Karadağ, T. “A Review of Memory Wall for Neuromorphic Computing”, *2025 IEEE 4th International Conference on Computing and Machine Intelligence (ICMI)*, pp. 1–6, 2025. doi:[10.1109/ICMI65310.2025.11141074](https://doi.org/10.1109/ICMI65310.2025.11141074)
- The remaining 23 peer-reviewed publications (38 in total) are listed at [Google Scholar](https://scholar.google.com/citations?user=IsmailCanDikmen).

Book Chapter

- [B1] Dikmen, İ. C., Karadağ, T. “Elektrikli Araç Batarya Yönetim Sistemleri ve Yeni Trendler”, [Electric Vehicle Battery Management Systems and New Trends] In: *Mühendislik Alanında Gelişmeler [Advances in Engineering]*, vol. 1, pp. 153–169, Platanus Publishing, ISBN 978-625-6971-84-4, 2023.

Manuscripts Under Review

- [U1] Dikmen, İ. C.[†] “Spiking-Front-End Mixture of Experts for Physics-Aware Anti-Jamming on a Baseband Digital Link”, *IEEE Transactions on Circuits and Systems for Video Technology*. · under review since Mar 2026
- [U2] Dikmen, İ. C.[†] “Genome-Heritable Neurons for Lossless Zero-Shot Inheritance in Spiking Networks”, *IEEE Transactions on Neural Networks and Learning Systems*. · under review since May 2026
- [U3] Dikmen, İ. C.[†] “Does Nature Know Fourier? Lessons from Echolocating Species for FFT-Free Neuromorphic Radar”, *Neurocomputing*. · under review since Jul 2026
- [U4] Dikmen, İ. C., Karadağ, T. “A Cross-Domain Double Dissociation of Neuron-Task Matching in Pure Spiking Networks”, *Neural Networks*. · submitted Jun 2026
- [U5] Dikmen, İ. C.[†] “nBMS, a Neuromorphic Battery Management System with a Silicon-Validated Spiking State-of-Charge Core for eVTOL Aircraft”, *IEEE Transactions on Aerospace and Electronic Systems*. · under review since Jul 2026

Preprints

- [R1] Dikmen, İ. C. “Spiking Neural Networks: A Tutorial on Models, Coding, and Training”, *Preprints.org*, 2026. doi:[10.20944/preprints202605.0827.v1](https://doi.org/10.20944/preprints202605.0827.v1) [Part 1 of a two-part series]
- [R2] Dikmen, İ. C. “Spiking Neural Networks: Mathematical Foundations”, *Preprints.org*, 2026. doi:[10.20944/preprints202605.1130.v1](https://doi.org/10.20944/preprints202605.1130.v1) [Part 2 of a two-part series]
- [R3] Yunusoğlu, A., Coşkun, T., Vishwamith, H., Işık, M., Dikmen, İ. C. “A Reconfigurable Framework for AI-FPGA Agent Integration and Acceleration”, *arXiv:2601.19263*, 2026. doi:[10.48550/arXiv.2601.19263](https://doi.org/10.48550/arXiv.2601.19263)
- [R4] Dikmen, İ. C. “Open-Loop Fidelity and Closed-Loop Validation with Computational Cost Analysis on Spiking Takagi-Sugeno Neuro-Fuzzy Control for Energy Constrained Systems”, *SSRN Working Paper 6709206*. doi:[10.2139/ssrn.6709206](https://doi.org/10.2139/ssrn.6709206)
- [R5] Işık, M., Vishwamith, H., Naoukin, J., Dikmen, İ. C. “HPCNeuroNet: A Neuromorphic Approach Merging SNN Temporal Dynamics with Transformer Attention for FPGA-Based Particle Physics”, *arXiv:2412.17571*, 2024. doi:[10.48550/arXiv.2412.17571](https://doi.org/10.48550/arXiv.2412.17571)
- [R6] Le, D., Yunusoğlu, A., Tiwari, K., Işık, M., Dikmen, İ. C. “Multimodal LLM for Intelligent Transportation Systems”, *arXiv:2412.11683*, 2024. doi:[10.48550/arXiv.2412.11683](https://doi.org/10.48550/arXiv.2412.11683)
- [R7] Pawlak, W. A., Işık, M., Le, D., Dikmen, İ. C. “Exploring Liquid Neural Networks on Loihi-2”, *arXiv:2407.20590*, 2024. doi:[10.48550/arXiv.2407.20590](https://doi.org/10.48550/arXiv.2407.20590)

PENDING PATENT APPLICATIONS

All filings with the Turkish Patent and Trademark Office (TÜRKPATENT).

- [P4] “Hazard-Function-Based Spiking Neuron Family, Liquid Extension, and N-ADA Heterogeneous Neuromorphic Processing System”, App. No. 2026/007632, filed 14 May 2026. Assignee: İstinye University. · *Pending*
- [P5] “Event-Driven Neuromorphic Sentinel-Based Heterogeneous System-on-Chip and Operating Method for Battery State Estimation”, App. No. 2026/004809, filed 31 Mar 2026. Assignee: İstinye University. · *Pending*
- [P6] “IoT-Based Medical Smart Transport Thermos”, App. No. 2022/021732, filed 30 Dec 2022. Assignee: İnönü University. · *Pending*
- [P7] “Hybrid and Electric Vehicle Battery Appraisal System and Method”, App. No. 2021/018933, filed 1 Dec 2021. Assignee: İnönü University. · *Pending*

TEACHING

Selected courses are listed; the complete course portfolio is available at candikmen.com

İstinye University — Signals and Systems, Electronics I.

Yıldız Technical University — Electric Vehicle Technologies*.

İnönü University — Selected Courses: Artificial Intelligence and Optimization Methods (graduate), Sensors and Transducers, Control Systems in Electric Vehicles, Automotive Electronics, and Advanced Sensor Technologies.

Malatya Turgut Özal University — Selected Courses: Automotive Electronics, Sensors and Transducers, Control Systems in Electric Vehicles, and Electric Vehicle Technologies*.

* Delivered under the university–industry cooperation programme jointly coordinated by the Ministry of Industry and Technology and the Council of Higher Education.

MENTORING AND OUTREACH

Graduate Thesis Supervision

- [M1] Yıldırım, N. “Elektrikli Araçlarda Kullanılan Farklı Kimyalara Sahip Lityum Tabanlı Pillerin Sağlık Durumlarının Analizi ve Modellenmesi”, [Analysis and Modelling of the State of Health of Lithium-Based Batteries with Different Chemistries Used in Electric Vehicles] *M.Sc. thesis, İnönü University, completed, 2025.* [Council of Higher Education Thesis Centre No. 10720357]

Pre-College Research Mentoring

- *Young Scientists High School Summer Research Internship, İstinye University — Neuromorphic Algorithms track. Research mentor, Jul 2026. Supervised three high school students through a two-week programme, each taking a spiking neural network study from problem formulation to a submitted conference manuscript. Two were submitted to the 10th International Artificial Intelligence and Data Processing Symposium (IDAP 2026) and one to the 30th IEEE High Performance Extreme Computing Conference (HPEC 2026).*
- DataMedX Hackathon 2 · İstinye University, clinical data analysis platform for oncology. Hackathon mentor, high school team, 2026.

Invited Talks

- Invited speaker, “Electric and Hydrogen Vehicles, Past to Present” (with T. Karadağ), 4th Automotive Summer Camp, Automotive Manufacturers Association (OSD), online, 2024.
- Invited speaker, “Transport Technologies on the Journey to Zero Emission”, Panel on hydrogen and electric vehicle technologies, 2nd Yenışehir Energy Days, Yenışehir Municipality, Mersin, January 2024.
- Invited speaker, “Intelligent Transportation Systems”, TÜBİTAK Science Talks, public science programme for K–12 students, online, 2023.

PROFESSIONAL SERVICE

Journal Reviewer

Reviewer for 12 international journals, including IEEE Transactions on Vehicular Technology, IEEE Transactions on Automation Science and Engineering, ISA Transactions, Scientific Reports, and Journal of Intelligent & Robotic Systems.

AWARDS AND HONOURS

- First Prize, Most Innovative Scientific-Based Project. RDCONF International Conference on Design and R&D, R&D and Innovation Awards, 2021, for an intelligent BMS for electric vehicles.
- NATO ISAF Medal. International Security Assistance Force, May 2012.